Language-Driven Control of a Humanoid Robot via the Model Context Protocol

Isak Greatrex · Linköping University · Master's thesis proposal · September 2026

Background

Agents built on large language models (LLMs) increasingly act by calling external tools [1–3]. The Model Context Protocol (MCP) has emerged as an open standard for this: a system exposes its capabilities as an MCP server, and any compatible agent can discover and use them [4–6]. In robotics, LLMs have been shown to plan multi-step tasks when restricted to choosing among a set of predefined robot skills [7] or when writing programs that compose robot control primitives [8, 9], and MCP has recently begun to be used as the interface between LLM agents and robotic devices [10] and humanoid robots [11].

Problem

A humanoid robot is not a software tool. LLMs have no built-in grounding in what a robot can physically do [7], agentic LLM control of real robots remains fragile, with non-deterministic behavior and instruction-following errors [12], and LLM response times are long and variable compared with the time scales on which a robot must react [10]. The key design question is therefore *what* the robot should expose to the agent: tools that are high-level enough to be used reliably by an LLM [9], bounded so that no call can drive the robot outside safe limits [11, 13–16], and backed by the robot's own sensing so that it can stop without waiting for the agent [10, 11, 17–19].

Aim and research questions

The aim is to design, implement, and evaluate an MCP server that lets any MCP-compatible LLM agent operate a SoftBank NAO V5 humanoid [20] through a small set of safe, high-level tools, such as moving, turning, looking around, stopping, and reporting the robot's state.

RQ1 How can a humanoid robot's capabilities be exposed as MCP tools so that LLM agents can use them effectively while all commands stay within safe bounds?

RQ2 How do different LLMs and agent applications compare in success rate, completion time and number of tool calls on the same multi-step natural-language tasks through this interface?

Approach

The work has two phases. First, the MCP server is developed and tested against a simulated NAO in the Webots simulator [21], where tasks can be repeated without risk to the hardware. Second, the same server is connected to a physical NAO V5 through NAOqi, the robot's native software framework [22]. Because NAOqi is used to control the robot in both phases, the MCP server and its tools stay the same; only the target changes from the simulated robot to the physical one. The system is evaluated by giving the same natural-language navigation and inspection tasks to several combinations of LLMs and agent applications (e.g. coding agents and desktop chat clients). Measures are task success, time to completion, and the number of tool calls per task.

Expected contributions

An open MCP server for the NAO, usable by any MCP-compatible agent, and an empirical evaluation of how well current LLM agents control a humanoid robot through a standard tool protocol, in simulation and on hardware.

References

[1] S. Yao et al. "ReAct: Synergizing Reasoning and Acting in Language Models". In: *International Conference on Learning Representations (ICLR)*. 2023.

[2] T. Schick et al. "Toolformer: Language Models Can Teach Themselves to Use Tools". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 36. 2023.

[3] Y. Qin et al. "Tool Learning with Foundation Models". In: *ACM Computing Surveys* 57.4 (2024).

[4] Anthropic. *Introducing the Model Context Protocol*. Nov. 25, 2024. `https://www.anthropic.com/news/model-context-protocol`.

[5] Model Context Protocol. *Model Context Protocol Specification, Version 2026-07-28*. July 28, 2026. `https://modelcontextprotocol.io/specification/2026-07-28`.

[6] X. Hou et al. "Model Context Protocol (MCP): Landscape, Security Threats, and Future Research Directions". In: *arXiv preprint arXiv:2503.23278* (2025).

[7] M. Ahn, A. Brohan, N. Brown, et al. "Do As I Can, Not As I Say: Grounding Language in Robotic Affordances". In: *Conference on Robot Learning (CoRL)*. 2022.

[8] J. Liang et al. "Code as Policies: Language Model Programs for Embodied Control". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2023.

[9] S. H. Vemprala et al. "ChatGPT for Robotics: Design Principles and Model Abilities". In: *IEEE Access* 12 (2024).

[10] Q. Guo et al. "Enabling Modular and Intelligent IoRT Systems: Integrating the Model Context Protocol for Semantic Decoupling at the Edge". In: *IEEE Internet of Things Magazine* 9.1 (2026).

[11] A. Bajrami, M. Elshamouty, and W. Kraus. "How Long Until Your Robot Ignores You? A Safety Benchmark for LLM Orchestrators in Human-Humanoid Collaboration". In: *arXiv preprint arXiv:2609.07288* (2026).

[12] O. Lima et al. "Agentic AI for Robot Control: Flexible but Still Fragile". In: *Proceedings of the AAAI Symposium Series*. Vol. 8. 1. 2026.

[13] L. Sha. "Using Simplicity to Control Complexity". In: *IEEE Software* 18.4 (2001).

[14] M. Alshiekh et al. “Safe Reinforcement Learning via Shielding”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 32. 1. 2018.
[15] A. Robey et al. “Jailbreaking LLM-Controlled Robots”. In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2025.
[16] A. Hundt et al. “LLM-Driven Robots Risk Enacting Discrimination, Violence, and Unlawful Actions”. In: *International Journal of Social Robotics* 17 (2025).
[17] E. Gat. “On Three-Layer Architectures”. In: *Artificial Intelligence and Mobile Robots: Case Studies of Successful Robot Systems*. Ed. by D. Kortenkamp, R. P. Bonasso, and R. Murphy. MIT Press, 1998.
[18] R. A. Brooks. “A Robust Layered Control System For A Mobile Robot”. In: *IEEE Journal of Robotics and Automation* 2.1 (1986).
[19] W. Huang et al. “Inner Monologue: Embodied Reasoning through Planning with Language Models”. In: *Proceedings of the 6th Conference on Robot Learning (CoRL)*. Vol. 205. Proceedings of Machine Learning Research. 2023.
[20] D. Gouaillier et al. “Mechatronic design of NAO humanoid”. In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2009.
[21] O. Michel. “Cyberbotics Ltd. Webots: professional mobile robot simulation”. In: *International Journal of Advanced Robotic Systems* 1.1 (2004).
[22] SoftBank Robotics. *ALMotion — NAOqi 2.5 Documentation*. 2017. `http://doc.aldebaran.com/2-5/naoqi/motion/almotion.html`.